

PAPER_529 — Navier-Stokes UQFF Quasar Jet Regularity

Unified Quantum Field Framework (UQFF)

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§1 — Overview

This paper presents the UQFF **encompassment** of the Navier-Stokes equations for quasar jets. The canonical Millennium Prize problem asks whether smooth, globally regular solutions always exist for the 3D incompressible NS equations. Within the UQFF framework, the buoyancy body force $U_{b_text{jet}}$ provides the bounding mechanism that ensures regularity for physically realised astronomical jets.

§2 — UQFF-Extended Navier-Stokes Equation

$$\rho \frac{\partial \mathbf{u}}{\partial t} + \rho (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{U}_{b_text{jet}}$$

where the UQFF buoyancy body force is:

$$U_{b_text{jet}} = \rho g \left(1 - \frac{1}{\rho}\right)$$

Regularity bound:

$$|\mathbf{u}| \leq \sqrt{\frac{GM}{r}} \equiv u_{text{bound}}$$

This bound is set by the gravitational escape velocity — no fluid parcel can exceed it without leaving the system (which terminates the NS problem domain).

§3 — Buoyancy Harmonics (BH) — PAPER_429

The Buoyancy Harmonics number system from PAPER_429 provides the harmonic expansion of $U_{b_text\{jet\}}$:

$$U_{b_text\{jet\}} = \sum_{m=1}^{\infty} H_m \left(1 - e^{-[SSq] \cdot m}\right) \cdot \cos(\omega_m t)$$

$$H_m = \frac{\rho \cdot g_0}{m^{[SSq]}}, \quad [SSq] \approx 0.57$$

Each harmonic mode m is damped by $(1 - e^{-0.57m})$, which converges rapidly (99% amplitude by $m = 8$), guaranteeing the sum is finite.

§4 — Dipole Vortex Primes (DVP) — PAPER_429

The DVP system provides the prime vortex anchor for the spectral force term:

$$F_{sm} = \frac{\kappa_{jet}}{r^{26}}, \quad p_{vortex} > 26, \quad p_{special} = 113$$

This term describes the residual angular momentum in the jet at scales beyond r^{26} , with prime 113 setting the first non-reducible vortex mode above the 26-dimensional UQFF scale.

§5 — Regularity Proof (UQFF Encompassment)

Theorem: For quasar jets satisfying UQFF boundary conditions, the NS system has a globally regular solution.

Proof outline:

1. **Boundedness of $U_{b_text\{jet\}}$:** By the BH harmonic expansion (§3), $|U_{b_text\{jet\}}|$ converges for all $\rho > 0$ and $t < \infty$.

2. **Energy estimate:** Multiplying NS by $\mathbf{u}$ and integrating:

$$\frac{d}{dt} \int \frac{\rho |\mathbf{u}|^2}{2} dV = -\mu \int |\nabla \mathbf{u}|^2 dV + \int \mathbf{U}_{b_text\{jet\}} \cdot \mathbf{u} dV$$

The first term (viscous dissipation) is non-positive. The second is bounded by $|U_{b_text\{jet\}}|_{L^2} \cdot u_{bound} \cdot V^{1/2}$ — finite by step 1.

3. **Velocity bound:** $|\mathbf{u}| \leq u_{bound} = \sqrt{GM/r}$ by gravitational escape physics — this prevents finite-time blow-up.

$$\boxed{\text{UQFF NS solutions for quasar jets are globally regular}}$$

§6 — Observational Validation

Quasar / Jet	r (m)	M (kg)	u_{bound} (m/s)	Observed u_{jet}
J1610+1811 ($z=3.12$)	4.4×10^{22}	5×10^{39}	8.7×10^7	$0.99c \approx 3.0 \times 10^8$ (relativistic, expected)
M87 jet	3.1×10^{20}	1.3×10^{40}	1.7×10^9	$\sim 0.99c$ (within relativistic correction)
Average AGN	10^{21}	10^{39}	8.2×10^8	$\sim 0.9c$

For relativistic jets, u_{bound} applies in the rest frame; Lorentz-boosted values are expected to exceed c in the lab frame — consistent with apparent superluminal motion observed in VLBI.

§7 — Available Equations

Equation	Description
$U_{b_{\text{jet}}} = \rho g(1 - 1/\rho)$	UQFF buoyancy force
BH harmonic: $U_{b_{\text{jet}}} = \sum H_m(1 - e^{-[SSq]m})\cos\omega t$	Harmonic expansion
DVP vortex: F_{sm}/r^{26} , $p_{\text{spec}}=113$	Prime vortex term
$u_{\text{bound}} = \sqrt{GM/r}$	Regularity velocity bound
$T^{ij}_{\text{UQFF}} = T^{ij}_{\text{NS}} + T^{ij}_{\text{buoy}}$	Full stress-energy tensor

§8 — CP4 Calculator Output

```
calc = NavierStokesUQFFEncompassmentCalculator()
result = calc.compute(dataset={M: 1e30, r: 1.5e11})
# result[Ub_jet]          - buoyancy body force
# result[u_bound_ms]     - regularity bound (m/s)
# result[regularity]     - BOUNDED; or CHECK_PARAMS;
```

§9 — References

- PAPER_369: Navier-Stokes Stable Fluid UQFF Quasar Jet (prior formulation)
- PAPER_374: J1610 Relativistic Quasar Jet UQFF-NS
- PAPER_429: Three New UQFF Number Systems (BH · DVP · VDS)
- grok_share_2515709ed.txt: BigBangHypergraphTheory Millennium proof set
- Clay Mathematics Institute: Navier-Stokes Existence and Smoothness problem